Supplementary Material: Residual TFT-EWC: Continual Learning for State-of-Health Forecasting and Reuse Screening of Heterogeneous Second-Life EV Batteries

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This supplement gives the preprocessing, the input correlations, the training curves and the full classification and regression metrics behind the paper. Every figure and table is generated by a script in the repository from committed results or from the public data; the script is named in each caption. SOH is relative to each cell's first valid capacity check. The additional results (Section S6) back the paper's main finding: of the models compared, Residual TFT-EWC is the only one that stays competitive both on the continual stream (Table SIII) and on unseen usage conditions (Table SV), while each traditional forecaster falls behind in one of the two settings. Section S7 gives the full results of the post hoc replay variant V2 on S7 and S10.

S1. Preprocessing

Preprocessing is the same for every model and uses no label information beyond the SOH history at the origin. It begins by verifying the raw files against their repository record or checksums; our parsing reproduces the processed S1 RPT capacities and the authors' capacities of all 843 S6 checks exactly. Validity rules then remove unusable checks: an S6 check is used only if its C/25 discharge starts from full charge and reaches the cut-off, which 4 of 847 checks fail (Fig. S1c), and in S1 RPT 2 (run at about 20 °C) is dropped.

Noise is reduced only where differentiation would amplify it. Incremental-capacity curves $-\mathrm{d}Q/\mathrm{d}V$ are computed on a 1 mV (S6) or 2 mV (S1) voltage grid after a Gaussian filter of the charge (5 mV in S6, 10 mV in S1); without this filter the measurement noise dominates the curve (Fig. S1b). Peak positions and heights are taken from the smoothed curve, and in S6 a peak at the edge of its window is set to missing. This is the only filtering; no other denoising or outlier removal is applied.

The ageing cycles and the model inputs are prepared last. Apache Spark reduces 9.0×10^8 per-second rows of the S6 ageing cycles to block inputs (mean, last value and slope of seven responses per 100-cycle block). Features never observed are dropped (e.g. the second IC window in S6), missing values are imputed with the training median and flagged, and inputs are standardised. Imputation, scaling and target scaling are fitted on the first task and then frozen.

S2. Input correlations

Fig. S3 gives the Spearman correlation of the S6 history and diagnostic inputs (F1) with each other and with the target. Ten of the 17 diagnostics are near-monotone functions of the SOH at the origin ($|\rho| \geq 0.9$), so they add little beyond the SOH history; the fade rate is the input most correlated with the target ($\rho = 0.8$). This is consistent with the ablation in the paper, where removing the diagnostics improved the TFT.

S3. Training curves

Each task is trained with Adam (batches of 64) for at most 200 epochs, with early stopping on the task's validation RMSE (patience 20); the weights of the best epoch are kept. Fig. S4 shows one stream of the residual TFT + EWC (horizons 1–3, first evaluation order, seed 0, tuned learning rate $3 \cdot 10^{-3}$ and $\lambda = 1$). The re-run reproduces the committed final average error of that stream exactly (0.464 pp). Validation RMSE reaches its minimum within 2–30 epochs on five tasks and after 86 epochs on task 6; on task 5 no epoch improved on the weights carried over from task 4, which were therefore kept.

S4. Full metric sets (P2)

Tables SI and SII give every regression and classification metric of the P2 forecasts (decision D30); Fig. S5 gives the S6 confusion matrices and Fig. S6 the error per held-out group.

S5. Ageing-cycle processing and first-life pretraining

Spark job. Between checks, every S6 cell logged each ageing cycle at about one sample per second: 827 files, 132 GB raw and 9.0×10^8 rows. Apache Spark 4.2 (PySpark, local mode) reduces them to block-level inputs: one map task per 100-cycle block derives, for every cycle, a validity flag and seven measured responses (end-of-charge and end-of-discharge voltage, constant-voltage hold time and final current, relaxation after charge and after discharge, internal resistance); Spark

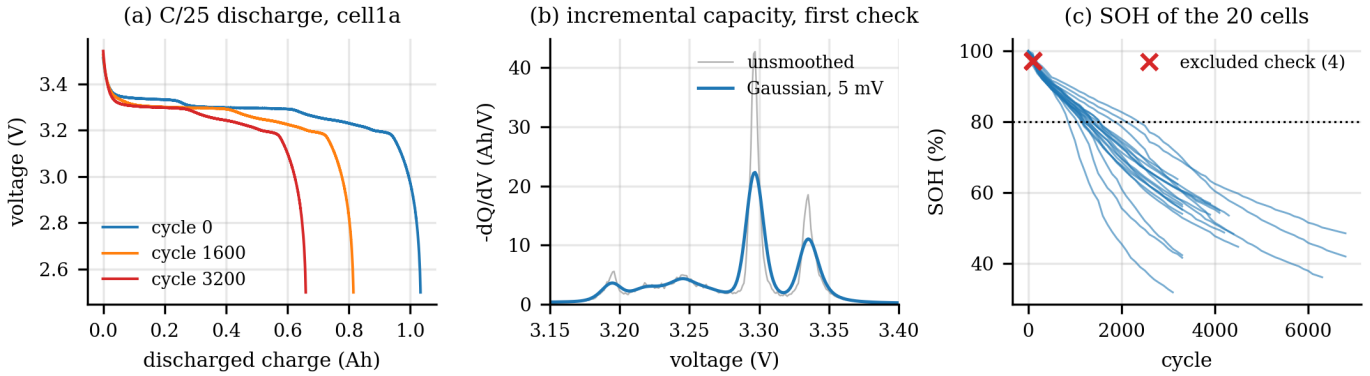


Fig. S1. S6 preprocessing (scripts/make_preprocessing_figures.py). (a) C/25 discharge of cell 1a at three ages. (b) Incremental capacity of its first check without and with the 5 mV Gaussian smoothing used for the IC features. (c) SOH of the 20 cells at every valid check; crosses: the four checks excluded by the validity rule, placed on the cell's curve at the cycle they were taken (they have no valid capacity). Dotted: 80% SOH.

TABLE SI

PROTOCOL P2, ALL REGRESSION METRICS OF THE FORECAST SOH (PP UNLESS STATED), MEAN OVER HELD-OUT GROUPS (THE RMSE COLUMN EQUALS TABLE SVII FOR THE BASELINES; MLP AND TFT ARE RE-RUNS ON ANOTHER PLATFORM, WITHIN 0.0052 PP OF IT). BIAS = MEAN(FORECAST − MEASURED): POSITIVE VALUES OVERESTIMATE SOH. NRMSE = RMSE / RANGE OF MEASURED SOH IN THE GROUP. R^2 IS ON THE SOH LEVEL, R^2_Δ ON THE SOH CHANGE SINCE THE ORIGIN (PERSISTENCE HAS $R^2_\Delta \leq 0$). SOH IS RELATIVE TO THE FIRST OBSERVED CHECK.

Data	Method	RMSE	MSE	MAE	MedAE	Bias	NRMSE	MAPE (%)	R^2	R^2_Δ
S1	Persistence	0.545	0.303	0.470	0.473	+0.430	0.161	0.481	0.672	-1.853
S1	Linear extrapolation	0.511	0.293	0.365	0.226	-0.014	0.148	0.374	0.690	-1.314
S1	Ridge, usage only	0.301	0.096	0.206	0.133	-0.038	0.088	0.212	0.897	0.181
S1	Ridge, F0	0.418	0.212	0.369	0.350	+0.205	0.118	0.379	0.798	-0.673
S1	Ridge, F1	0.320	0.115	0.266	0.241	+0.065	0.092	0.273	0.881	0.083
S1	Gradient boosting, F1	0.219	0.053	0.148	0.102	-0.021	0.064	0.152	0.944	0.543
S1	MLP	0.285	0.087	0.211	0.162	+0.009	0.083	0.216	0.907	0.260
S1	TFT	0.291	0.090	0.210	0.149	+0.027	0.085	0.215	0.904	0.216
S6	Persistence	2.817	8.389	2.451	2.225	+2.451	0.056	3.539	0.956	-3.160
S6	Linear extrapolation	1.372	1.946	0.738	0.333	+0.050	0.028	0.973	0.988	-0.050
S6	Ridge, usage only	1.177	1.607	0.864	0.671	-0.003	0.023	1.292	0.993	0.275
S6	Ridge, F0	0.859	0.905	0.613	0.439	+0.028	0.016	0.893	0.996	0.614
S6	Ridge, F1	1.137	1.767	0.749	0.550	+0.013	0.022	1.130	0.991	0.221
S6	Gradient boosting, F1	0.710	0.725	0.488	0.321	+0.097	0.013	0.710	0.997	0.736
S6	MLP	0.826	1.018	0.608	0.444	+0.220	0.015	0.914	0.996	0.656
S6	TFT	0.868	1.151	0.621	0.424	+0.235	0.016	0.928	0.996	0.622

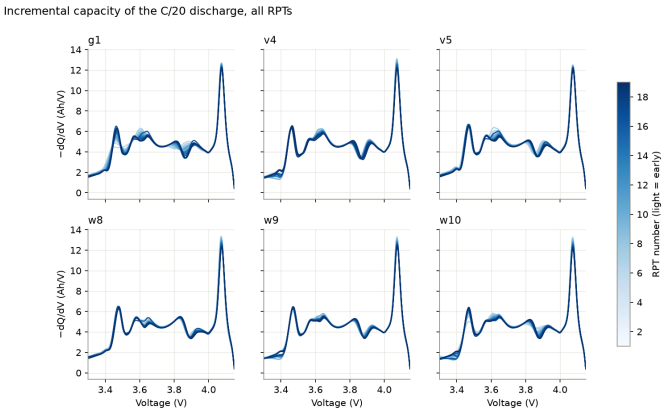


Fig. S2. S1: incremental-capacity curves of the C/20 RPTs (10 mV smoothing), notebook 11.

SQL groups the cached per-cycle table (82,529 cycles) by cell and block into the mean, last value and slope of each response (21 inputs), and each block joins the origin at the check that ends it. Charge, energy and current are excluded because the protocol scales them with the previous capacity (within-cell $r = 0.97$–0.999 with the previous check's capacity). The run takes 20 min on six cores and matches an independent pandas implementation on all 827 blocks (largest difference 2×10^{-13}).

First-life pretraining (P3). The pool provides 36,711 samples, against 2,409 in S6, and validation selected EWC $\lambda = 0.01$ for the pretrained TFT. Under the replication rule, EWC and fine-tuning are practically equivalent for the pretrained TFT (differences of 0.015–0.017 pp). The S6 stream does not degrade the pool test error (TFT + EWC: 3.280 to 3.148 pp), but on that split the pretrained models are also less accurate than extrapolation (3.001 pp): pretraining does not learn a first-life model worth transferring.

Attention. The TFT's attention over the four previous origins is nearly uniform (0.24–0.26 per lag): it makes little use of its history.

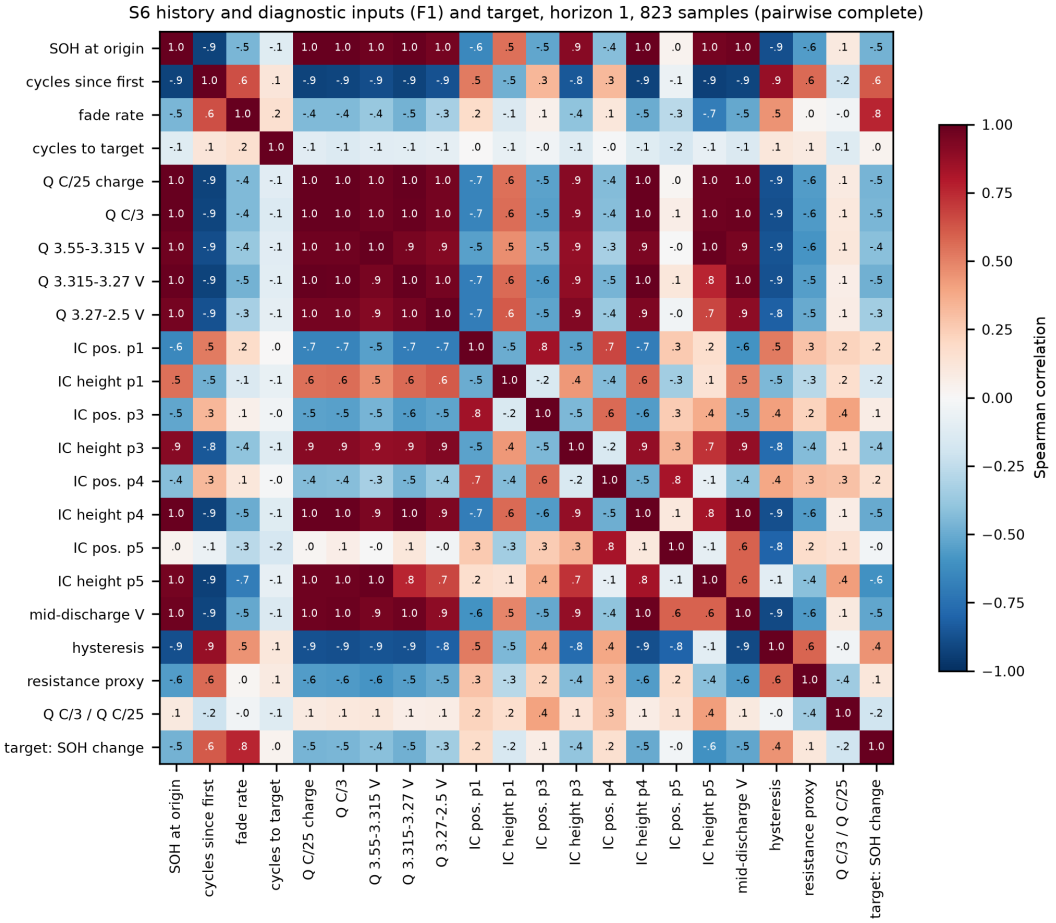


Fig. S3. Spearman correlation of the S6 F1 inputs and the target Δ (horizon 1, pairwise complete samples; the static test conditions are omitted). scripts/make_preprocessing_figures.py.

S6. Additional results

Tables SIII and SVI and Figs. S7, S8, S9 and S10 belong to Sections IV-D to IV-G of the paper and were moved here for space.

Reproducibility of the comparisons (Section IV-B). P1 and P2 were re-run in a pinned environment on data rebuilt from the raw archives. Every baseline, every P2 result and the ranking of methods reproduced; the neural models moved by at most 0.059 pp. On S1, five Holm-corrected comparisons with TFT + EWC crossed $p = 0.05$. On S6, validation selected EWC $\lambda = 1000$ instead of 100 (a 0.008 pp tie), and the error advantage of TFT + EWC over fine-tuning moved from $p_{\text{Holm}} = 0.005$ to 0.060, while its reduction of forgetting moved from 0.050 to 0.008. These findings motivated the ten tuning orders, the one-standard-error rule and the replication rule used from P3 on. A D30 re-run of P2 on another platform reproduced every baseline exactly and moved the MLP and TFT by at most 0.027 pp per held-out group: neural training is deterministic per platform, not across platforms.

Reuse classes in the D30 re-run of P2. On S6 the learned models assign the measured reuse class to 95–97% of samples (MCC 0.90–0.93), against 94% (0.88) for linear extrapolation and 86% (0.72) for persistence, and all errors but one (persistence) are between neighbouring classes; the end-of-first-life decision does not separate the methods (AUC $\geq$0.998), as 69% of S6 samples are below 80%.

Reuse classes on unseen S6 conditions (Section IV-F of the paper). The proposed model puts 95.5% of held-out samples in their measured class, against 93.9% for extrapolation and 85.8% for persistence, while ridge (96.5%) and gradient boosting (97.7%) do better (Table SVI, Fig. S9). Every error is off by one class. The most common one is also the risky one for reuse: a sample measured as C1 but forecast as C0 (65 of 182 for extrapolation, 42 for the proposed model, 7 for gradient boosting). The pretrained TFT reaches 0.509/0.591 pp against 0.401/0.415 pp from scratch (significant only in the replicate set).

Intervals on unseen conditions. Intervals (Fig. S8) calibrated on the pooled validation rows of the training conditions undercover on the held-out one, for the TFT (pooled 0.66–0.68) and for extrapolation (0.74–0.76) alike. Calibrating extrapolation, which has no fitted parameters, on all origins of those

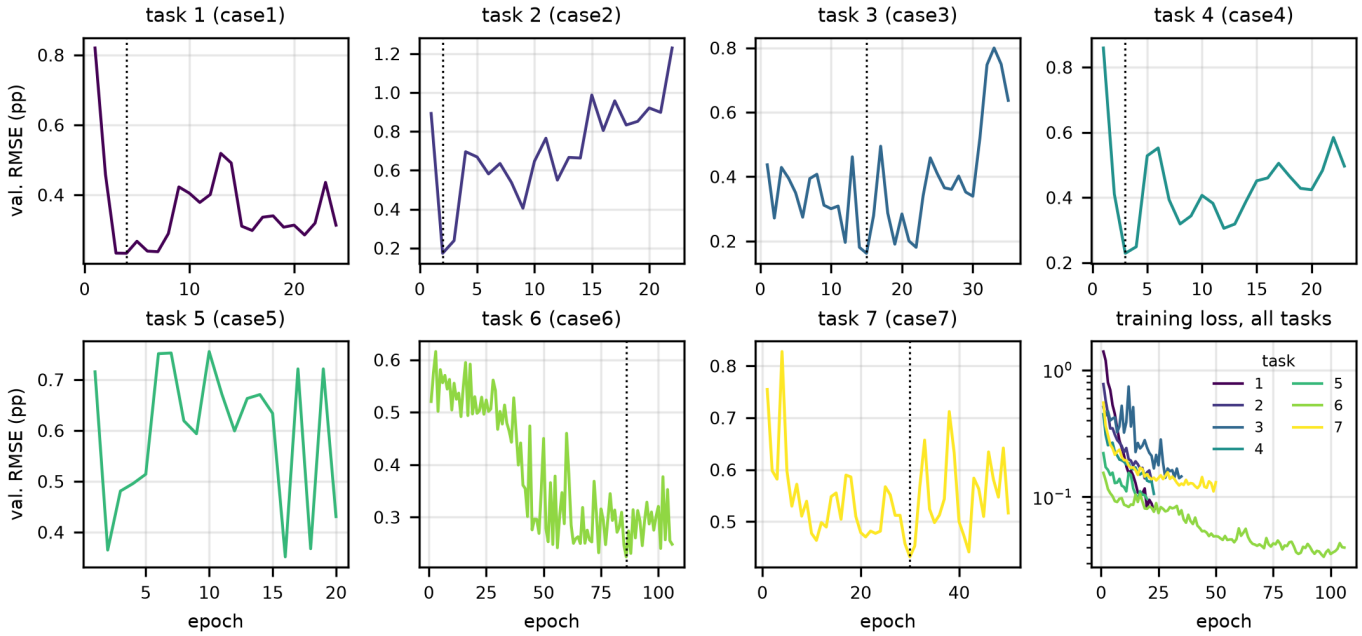


Fig. S4. Residual TFT + EWC on the S6 stream: validation RMSE per epoch for each task (dotted: early-stopping epoch) and training loss (scaled MSE plus EWC penalty) of all tasks. scripts/run_epoch_curves.py, results/epoch_curves_d23_e1.json.

TABLE SII

PROTOCOL P2, REUSE CLASS (C0–C3, D10 CUT-POINTS) AND END-OF-FIRST-LIFE DECISION (SOH $\leq$ 80, POSITIVE) OF THE FORECAST SOH AGAINST THE MEASURED SOH, ALL HELD-OUT SAMPLES POOLED (MEAN OVER SEEDS). MACRO AVERAGES RUN OVER THE MEASURED CLASSES. AUC USES THE FORECAST SOH AS THE SCORE (POINT FORECASTS; NO CLASS PROBABILITIES, SO NO LOG LOSS). MEASURED CLASS COUNTS OF ALL SAMPLES: S1: C0 279, C1 0, C2 0, C3 0; S6: C0 52, C1 182, C2 516, C3 1659. – : UNDEFINED (ONE CLASS ONLY).

Data	Method	Acc.	Bal. acc.	Prec.	Rec.	Spec.	F1	MCC	κ	κ_w	EoL Prec.	EoL Rec.	EoL Spec.	EoL F1	EoL MCC	EoL AUC
S1	Persistence	1.00	1.00	1.00	1.00	–	1.00	–	–	–	–	–	1.00	–	–	–
S1	Linear extrapolation	0.98	0.98	1.00	0.98	–	0.99	–	0.00	0.00	–	–	1.00	–	–	–
S1	Ridge, usage only	1.00	1.00	1.00	1.00	–	1.00	–	0.00	0.00	–	–	1.00	–	–	–
S1	Ridge, F0	1.00	1.00	1.00	1.00	–	1.00	–	–	–	–	–	1.00	–	–	–
S1	Ridge, F1	1.00	1.00	1.00	1.00	–	1.00	–	–	–	–	–	1.00	–	–	–
S1	Gradient boosting, F1	0.99	0.99	1.00	0.99	–	0.99	–	0.00	0.00	–	–	1.00	–	–	–
S1	MLP	1.00	1.00	1.00	1.00	–	1.00	–	0.00	0.00	–	–	1.00	–	–	–
S1	TFT	1.00	1.00	1.00	1.00	–	1.00	–	0.00	0.00	–	–	1.00	–	–	–
S6	Persistence	0.86	0.78	0.62	0.78	0.96	0.66	0.72	0.72	0.81	1.00	0.93	1.00	0.96	0.89	1.00
S6	Linear extrapolation	0.94	0.84	0.81	0.84	0.98	0.78	0.88	0.87	0.91	1.00	0.98	0.99	0.99	0.96	1.00
S6	Ridge, usage only	0.95	0.91	0.83	0.91	0.98	0.85	0.90	0.90	0.93	0.99	0.98	0.98	0.99	0.96	1.00
S6	Ridge, F0	0.96	0.88	0.89	0.88	0.99	0.88	0.92	0.92	0.94	1.00	0.98	0.99	0.99	0.97	1.00
S6	Ridge, F1	0.96	0.84	0.88	0.84	0.98	0.85	0.91	0.91	0.93	0.99	0.99	0.97	0.99	0.96	1.00
S6	Gradient boosting, F1	0.97	0.89	0.89	0.89	0.99	0.89	0.93	0.93	0.95	0.99	0.99	0.98	0.99	0.97	1.00
S6	MLP	0.96	0.86	0.88	0.86	0.99	0.87	0.92	0.92	0.94	0.99	0.99	0.98	0.99	0.97	1.00
S6	TFT	0.96	0.88	0.89	0.88	0.99	0.88	0.92	0.92	0.94	0.99	0.98	0.99	0.99	0.96	1.00

conditions instead of their mid-life validation rows restores 0.89–0.90: the calibration rows caused the shortfall, not only the model. With leave-one-condition-out calibration, gradient-boosting intervals reach the registered threshold of 0.85 (0.90 in both horizon sets) with the lowest mean interval scores (3.2 and 6.7 pp, against 7.1 and 11.8 pp for calibrated extrapolation and 6.1–6.3 and 10.0–10.3 pp for the TFT), lower than calibrated extrapolation's on six of seven conditions (significant only for horizons 1–3, $p = 0.047$, against $p = 0.30$, so not established). Coverage holds over conditions, not for each: on the 100% depth-of-discharge condition, outside the range of the others, it is 0.48 and 0.38.

Continual stream (Table SIII). Learning the residual makes the TFT the most consistent continual learner: for horizons 1–3 it is level with linear extrapolation and with the best literature backbones, and for horizons 1–6 it is ahead of the residual BiLSTM and TCN (replicated). With EWC its forgetting over six horizons is about a third of fine-tuning's (0.09 against 0.22–0.28 pp) and lower than that of every other backbone (0.13–0.35 pp). Gradient boosting, SVR and GPR, although fitted on all tasks at once, are less accurate on average.

Unseen conditions (Table SV). The proposed model is more accurate than linear extrapolation and persistence on all origins and on origins with a fade rate. Gradient boosting and the

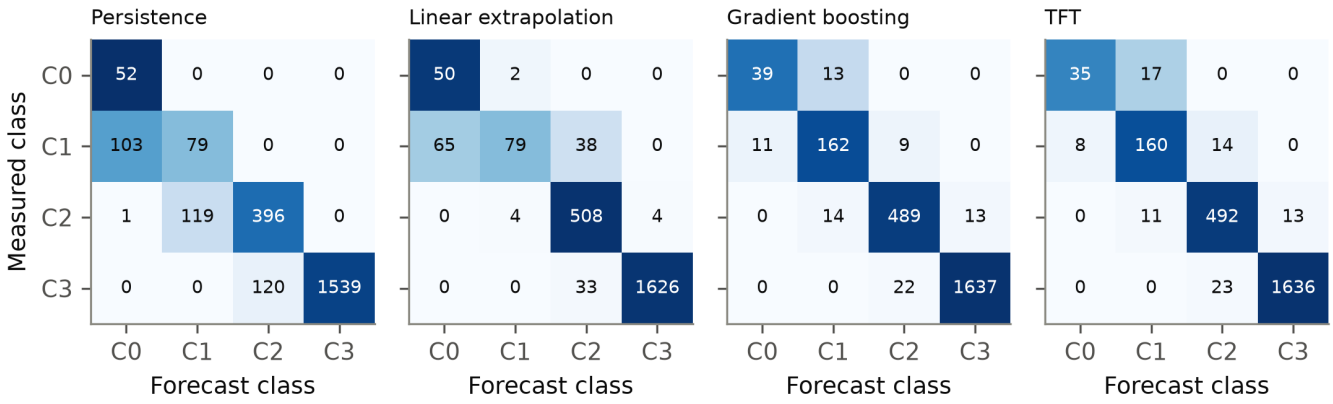


Fig. S5. S6, P2: reuse-class confusion matrices (counts; shading: row share), seed 0. scripts/make_d30_outputs.py.

TABLE SIII
LEARNED RESIDUAL, SHIFT-ADAPTIVE EWC AND LITERATURE BASELINES ON S6 (STREAM FROM SCRATCH): FINAL AVERAGE TEST RMSE (PP), MEAN OVER 50 RUNS, MAIN / REPLICATE SEED SETS, AND FORGETTING (PP) FOR HORIZONS 1–6. BELOW THE LINE: SINGLE FITS ON ALL TASKS (NOT CONTINUAL) AND EXTRAPOLATION.

Method	E1: h=1–3	E2: h=1–6	Forgetting, E2
TFT + EWC, residual	**0.339 / 0.314**	**0.561 / 0.534**	**0.093 / 0.086**
TFT + SA-EWC, residual	0.343 / 0.321	0.548 / 0.564	0.066 / 0.084
TFT fine-tuning, residual	0.347 / 0.337	0.673 / 0.612	0.282 / 0.220
TFT + EWC, direct	0.504 / 0.487	0.802 / 0.835	0.135 / 0.166
MLP + EWC, residual	0.374 / 0.366	0.659 / 0.694	0.205 / 0.225
BiLSTM + EWC, residual	0.328 / 0.370	0.669 / 0.765	0.229 / 0.318
Transformer + EWC, residual	0.360 / 0.366	0.609 / 0.705	0.129 / 0.198
TCN + EWC, residual	0.466 / 0.408	0.812 / 0.756	0.345 / 0.268
Linear extrapolation	0.343	0.485	–
Gradient boosting	0.451	0.898	–
SVR, residual	0.429	0.616	–
GPR, direct	1.374	2.073	–

jointly trained TFT are more accurate still, but both are fitted on all six training conditions at once; under the continual constraint, gradient boosting is within 0.08 pp (see below). The table lists the shrunk residual TFT + EWC; the raw correction gives 0.88/0.90, 0.79/0.80, 1.59/1.62 and 1.46/1.49 pp in the same columns.

Shift-adaptive EWC and literature backbones (Sections III-E and IV-D of the paper). The models fitted once see all tasks at the same time, and the direct literature models were run for horizons 1–3 only. The second registered experiment of the paper lets the EWC strength follow how far the data shift between groups (shift-adaptive EWC, SA-EWC): when task t starts, the EWC penalty (Eq. (4) of the paper) of each finished task j is multiplied by $d_{j,t} = \left((\|\mu_t - \mu_j\|^2 + \|\sigma_t - \sigma_j\|^2)/D\right)^{1/2}$, the per-dimension 2-Wasserstein distance between diagonal Gaussians fitted to the two tasks' preprocessed training inputs, with λ_0 tuned like λ. BiLSTM, TCN and Transformer-encoder continual learners (TFT input window, within 20% of its parameters) and GPR, random forest and SVR fitted once are run in residual and direct form. The second registered criterion, for SA-EWC (Table SIII), was *not* met either: although the shift weights varied (0.05–1.14), SA-EWC and a fixed λ are practically equivalent in final error and forgetting (differences of at most 0.03 pp). For horizons 1–3, the BiLSTM and Transformer with EWC are practically equivalent to the residual TFT; for 1–6 it beats the residual BiLSTM and TCN (replicated) but not the Transformer. Every model fitted once on all tasks is worse than the residual TFT + SA-EWC (replicated), except the residual SVR for horizons 1–6.

Many unseen conditions (S7 and S10, Section IV-G of the paper). Both tests were registered before any model was fitted on the dataset (D34, D36) and were motivated by the wish to find a setting where the proposed model beats the traditional forecasters. S10 (Stroebl et al., 2024; Samsung INR21700-50E) has 279 cells in two stages; condition = (stage, test point), since test-point names repeat across stages. Its label is the discharge capacity of the last complete 1C + CV discharge of each 23 °C check-up; check-ups were dropped for surface temperature outside 18–28 °C at its start (20), dates out of order (13), no complete discharge (3) or no usable date (3), and cells with fewer than five valid check-ups (66 calendar cells have only the entry and final tests) are not used. On S7 the registered model shrinks at every origin; at H^- origins this pulls the forecast towards persistence and made it worse than the raw model (E1: 6.66 against 4.43 pp on H^-), so the S10 test shrinks only at H^+ origins. On S10 this rule is practically equivalent to shrinking everywhere, and better than the raw model for horizons 1–3 only (-0.07 pp). Against gradient boosting refitted on the latest condition, the S10 model is better at both H^+ (-0.94, -1.07 pp) and H^- origins (-0.61, -0.31 pp; E1, E2; replicated). On S7 the advantage is at H^+ (-2.2 to -2.3 pp) while the model is worse at H^- ($+1.4$ to $+2.1$ pp). Gradient boosting with every condition stored has a larger error than the proposed model on S10 calendar conditions (E2 mean 12.5 against 1.4 pp) and a smaller one

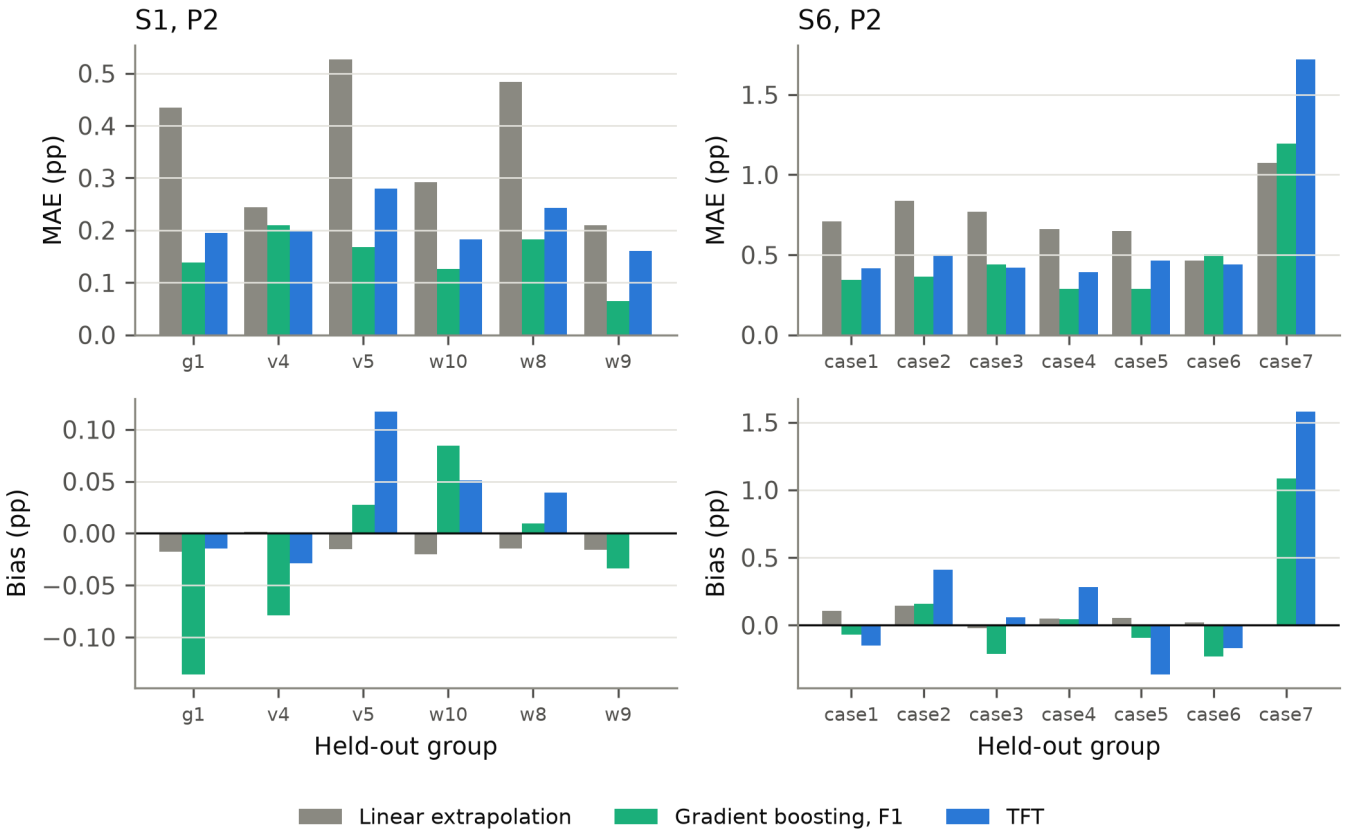


Fig. S6. P2: error per held-out group. scripts/make_d30_outputs.py.

on cyclic conditions (1.25 against 1.61 pp). In the registered rule the direction comes from the mean difference, so for S10, horizons 1–3, the test against it reads "better" although the median difference (+0.17 pp) and 52 of 64 conditions favour gradient boosting; we do not count it as a win. The tuning chose learning rates of 10^{-5} to $3 \cdot 10^{-3}$ and EWC strengths of 10^{-6} to 1, mostly far below the S6 values, so in most folds EWC adds little penalty. Pooled 90% coverage of the intervals is 0.71/0.57 (S7) and 0.74/0.43 (S10) for E1/E2, below the 0.85 the registration required.

Refit and tracking (Section IV-F of the paper). In the sixth experiment on unseen conditions the held-out condition arrives over time: as each target is measured, gradient boosting is refitted with it and the half-width moves by $c\,q_h(e-a)$, with $e = 1$ for a miss (quantile tracking; c chosen on the training conditions). Refitting gradient boosting as each check of the held-out condition arrives, and adjusting the interval half-width after every hit or miss, lowers its mean 90% interval score by 29%: from 3.22 to 2.29 pp for h=1–3 (lower on six of seven conditions, $p = 0.047$) and from 6.67 to 4.74 pp for h=1–6 (seven of seven, $p = 0.016$), with pooled coverage 0.90–0.91. On the 100% depth-of-discharge condition coverage rises from 0.48/0.38 to 0.72/0.58, still short of the registered 0.80. Of the held-out intervals, 87–93% then lie within one reuse class, and 99.2–99.7% of those name the right class.

Gradient boosting under the continual constraint (D32). Most of gradient boosting's advantage on unseen conditions comes from storing every training condition. The headline comparison of the paper (its Table III) therefore also trains gradient boosting like the proposed model: the six training conditions arrive one at a time and earlier conditions' data are not kept. Under this constraint the two are close. Refitted on the latest condition only, gradient boosting reaches 0.86 and 1.59 pp (horizons 1–3 and 1–6), against 0.92/0.94 and 1.62/1.65 pp for the proposed model (slightly better for 1–3, $+0.06/+0.08$ pp, replicated; not distinguishable for 1–6), whereas with all six conditions stored it reaches 0.54 and 1.05 pp. Table SIV is the first P1 benchmark (Section IV-A of the paper). Warm-started condition by condition (200 more trees per condition), it reaches 2.65 and 3.49 pp, and the proposed model is better by 1.7–1.9 pp (replicated). This gap is mostly an artefact: scikit-learn keeps the bin edges of the first condition when warm-starting, and every usage input is constant within a condition, so the later trees cannot use them. We therefore do not count it as evidence against continual gradient boosting in general.

S7. Replay ensemble V2 on S7 and S10 (D38)

This section gives the full results of the bounded-memory variant V2 (Section IV-H and Table VIII of the paper),

TABLE SIV

PROTOCOL P1 (CONTINUAL STREAM). FINAL AVERAGE TEST RMSE OF SOH(k+h) IN PP (MEAN ± S.D. OVER 10 ORDERS × 5 SEEDS FOR CONTINUAL METHODS, OVER 5 SEEDS FOR JOINT TRAINING), SKILL RELATIVE TO PERSISTENCE ($1 - \mathrm{RMSE}/\mathrm{RMSE}_{\mathrm{persistence}}$) AND FORGETTING (PP). †UPPER BOUND, TRAINED ON ALL TASKS AT ONCE.

Method	S1: six NMC second-life cells RMSE	Skill	Forgetting	S6: 20 LFP cells, seven usage conditions RMSE	Skill	Forgetting
Persistence	0.632	0.00	–	1.644	0.00	–
Linear extrapolation	0.726	-0.15	–	0.343	0.79	–
Ridge, usage only	0.450	0.29	–	1.274	0.22	–
Ridge, F0	0.678	-0.07	–	0.801	0.51	–
Ridge, F1	0.491	0.22	–	0.605	0.63	–
Gradient boosting, F1	0.466	0.26	–	0.372	0.77	–
MLP fine-tuning	0.492 ± 0.029	0.22	0.030	0.531 ± 0.151	0.68	0.207
MLP replay	0.475 ± 0.021	0.25	0.025	0.597 ± 0.178	0.64	0.212
MLP + EWC	0.497 ± 0.033	0.21	0.040	0.540 ± 0.176	0.67	0.212
MLP joint†	0.500 ± 0.016	0.21	–	0.414 ± 0.037	0.75	–
TFT fine-tuning	0.467 ± 0.014	0.26	0.026	0.883 ± 0.538	0.46	0.461
TFT + EWC	0.462 ± 0.019	0.27	0.016	0.677 ± 0.261	0.59	0.274
TFT + EWC, F0	0.472 ± 0.020	0.25	0.019	0.621 ± 0.257	0.62	0.176
TFT joint†	0.455 ± 0.017	0.28	–	0.427 ± 0.049	0.74	–

TABLE SV

UNSEEN S6 USAGE CONDITION (P2, SEVEN HELD-OUT CONDITIONS): TEST RMSE (PP), MEAN OVER FOLDS, ON ALL ORIGINS AND ON ORIGINS WITH A FADE RATE (H^+). TFT ROWS: MEAN OVER 50 RUNS, MAIN / REPLICATE SEED SETS; OTHER ROWS: SINGLE DETERMINISTIC FITS.

Method	E1: h=1–3 all	E1: h=1–3 H^+	E2: h=1–6 all	E2: h=1–6 H^+
TFT + EWC, residual, shrunk	0.92 / 0.94	0.74 / 0.74	1.62 / 1.65	1.42 / 1.44
TFT, residual, joint	0.57 / 0.58	0.54 / 0.55	1.21 / 1.18	1.22 / 1.19
Gradient boosting	**0.54**	**0.51**	**1.05**	**1.05**
Ridge	0.77	0.69	1.30	1.26
Linear extrapolation	1.37	0.86	2.30	1.62
Persistence	2.82	2.66	4.90	4.73

TABLE SVI

REUSE CLASS OF THE FORECAST SOH ON UNSEEN S6 CONDITIONS (P2, HORIZONS 1–3, SEVEN HELD-OUT CONDITIONS POOLED; MEASURED C0 52, C1 182, C2 516, C3 1659). SENS., PREC.: MACRO RECALL AND PRECISION. PROPOSED MODEL: MEAN AND S.D. OVER 50 RUNS; OTHERS: SINGLE FITS. BOLD: BEST.

Approach	Acc.	Sens.	Prec.	F1	MCC
Persistence	0.858	0.782	0.625	0.661	0.723
Linear extrapolation	0.939	0.840	0.810	0.777	0.876
Ridge	0.965	0.857	0.861	0.858	0.927
Gradient boosting	**0.977**	**0.924**	**0.926**	**0.925**	**0.952**
Residual TFT-EWC (proposed)	0.955	0.880	0.840	0.840	0.906
s.d. over 50 runs	±0.005	±0.017	±0.016	±0.021	±0.011

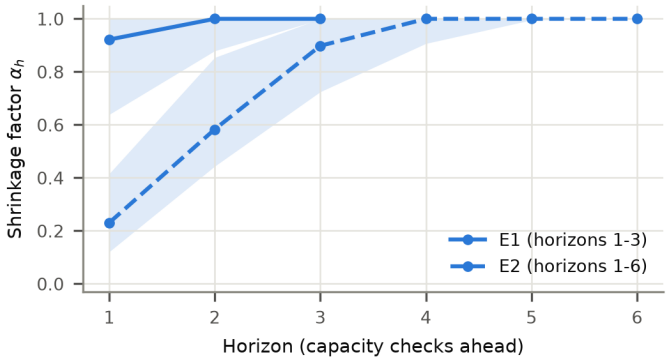


Fig. S7. S6 stream: shrinkage factor α_h of the TFT correction per horizon after the last task (median and interquartile range over 50 runs, seeds 0–4); $\alpha_h = 0$ is linear extrapolation.

written from `results/forecasting_d38_s10_e1.json`, `_s10_e2`, `_s7_e1` and `_s7_e2` and the D38 entry of `docs/DECISIONS.md`.

Provenance. D38 was registered after every earlier result and after the S7 and S10 test results had been read per condition; its protocol was fixed, and its open defaults accepted, before any D38 code ran on real data. Its test conditions, folds, evaluation orders and horizon sets are those of D34 (S7) and D36 (S10). It is therefore a development test with a protocol fixed in advance, not a confirmatory test; a confirmatory test needs a dataset not used so far. Only the replay learning rate was tuned, per fold, on the seven tuning conditions with the one-standard-error and edge rules; the selected values (folds 0–4) are $3 \cdot 10^{-3}$, 10^{-2}, 10^{-3}, 10^{-2}, $3 \cdot 10^{-3}$ (S10, h=1–3), 10^{-3}, $3 \cdot 10^{-3}$, 10^{-2}, 10^{-3}, 10^{-3} (S10, h=1–6), 10^{-3}, 10^{-2}, $3 \cdot 10^{-3}$, 10^{-2}, 10^{-3} (S7, h=1–3) and $3 \cdot 10^{-3}$, 10^{-4}, 10^{-3}, $3 \cdot 10^{-3}$, $3 \cdot 10^{-4}$ (S7, h=1–6). The S10 results were run and collected at commit `a093b37` on a clean tree. The S7 results were run with the code at `e9e66b2`; the run was interrupted and resumed from its checkpoints several times (no setting or seed was changed and no part was re-run to change an outcome), and both S7 results files record `git_dirty: true` because uncommitted edits to `paper/` were restored in the working tree during the run; no file outside `paper/` differed from `e9e66b2`. Environment: Python 3.11.15, NumPy 2.4.6, pandas 3.0.6, scikit-learn 1.9.1, PyTorch 2.14.1 (CPU). All 70 parts of each dataset finished; there are 64 (S10) and 60 (S7) test conditions and 6,495 and 11,093 (S10) and 10,251 and 18,762 (S7) samples for horizons 1–3 and 1–6. Every run also re-ran the unchanged EWC model (arm A0) as a reproduction check: it matched the committed D36 and D34 values within $5.3 \cdot 10^{-15}$ pp per condition and seed set, and gradient boosting,

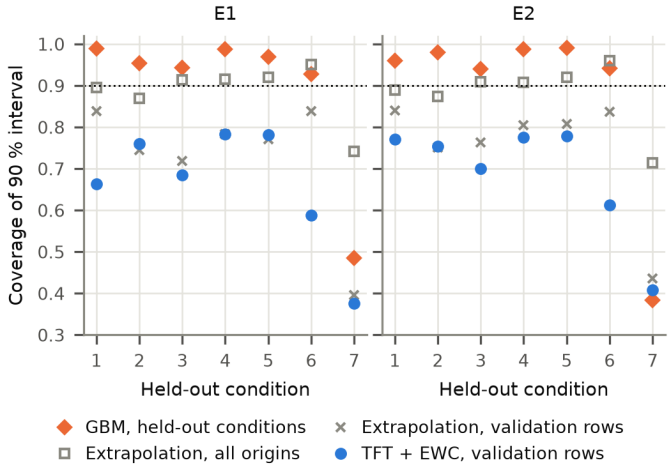


Fig. S8. Unseen S6 conditions: coverage of 90% conformal intervals per held-out condition, by forecaster and calibration rows. TFT: mean over 50 runs (seeds 0–4). Dotted: nominal level.

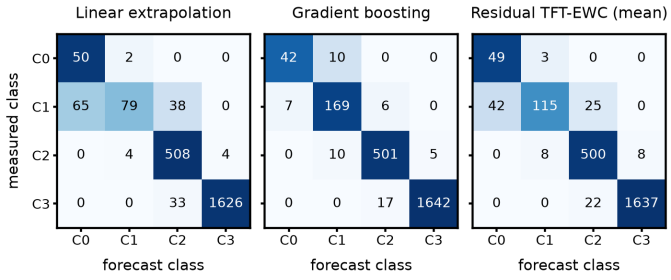


Fig. S9. Reuse-class confusion on unseen S6 conditions (Table SVI): sample counts (proposed model: mean over 50 runs); shading: share of the measured class. scripts/make_d26_class_outputs.py.

extrapolation, persistence and ridge matched exactly.

Arms. R32 is the replay model with 32 training and 8 validation rows stored per condition and R8 the one with 8 and 2 (at R32's learning rate); V2 is the calibrated R32 ensemble. GB-ER32 and GB-ER8 are gradient boosting fitted on the last condition of each evaluation order plus the rows a seed-0 replay model stores at that point. The joint residual TFT values are read from the committed D34 and D36 files, not refitted.

Accuracy (Tables SIX–SXI). On S10 V2 answers Q1–Q5 positively and on S7 Q3–Q5; on S7 gradient boosting with all conditions stored (Q1) and GB-ER32 (Q2) are better, replicated, with medians of the same sign as the means. On S10 the Q1 and Q2 means are dominated by the conditions where gradient boosting fails; without the condition `s2_z02`, named in advance because it dominates the S10 means, the means (main/replicate) are 0.536/0.539 (V2), 0.547/0.539 (joint TFT), 0.766/0.764 (earlier model), 1.219 (GB, all stored), 1.363 (GB-ER32) and 0.939 pp (extrapolation) for horizons 1–3, and 0.776/0.767, 0.735/0.796, 1.289/1.298, 4.695, 5.044 and 1.634 pp for 1–6. Experience replay accounts for most of the gain over the EWC model; the seed ensemble adds a smaller part; calibration leaves accuracy unchanged

TABLE SVII

PROTOCOL P2 (UNSEEN BATTERY IN S1, UNSEEN USAGE CONDITION IN S6): MEAN TEST RMSE (PP) OVER HELD-OUT GROUPS, RANGE OVER GROUPS. REUSE-CLASS METRICS OF THE SAME FORECASTS: TABLE IV OF THE PAPER.

Data	Method	RMSE	Range
S1	Persistence	0.545	0.436–0.688
S1	Linear extrapolation	0.511	0.284–0.725
S1	Ridge, usage only	0.301	0.172–0.409
S1	Ridge, F0	0.418	0.158–0.779
S1	Ridge, F1	0.320	0.161–0.477
S1	Gradient boosting, F1	0.219	0.091–0.299
S1	MLP	0.285	0.149–0.389
S1	TFT	0.292	0.199–0.377
S6	Persistence	2.817	2.025–4.327
S6	Linear extrapolation	1.372	0.856–1.669
S6	Ridge, usage only	1.177	0.870–2.288
S6	Ridge, F0	0.859	0.598–1.836
S6	Ridge, F1	1.137	0.588–2.188
S6	Gradient boosting, F1	0.710	0.395–1.839
S6	MLP	0.821	0.476–2.225
S6	TFT	0.868	0.551–2.390

and provides the intervals. Memory size makes no difference on S10, while on S7 R8 is better than R32 for horizons 1–6 (replicated). GB-ER32 is worse than GB-ER8 on S10 on average; this is reported, not explained. Counted per condition (main seeds), the joint TFT is best on 19 and 29 of the 64 S10 conditions and the four replay-ensemble variants together on 29 and 25 (horizons 1–3 and 1–6); on S7, gradient boosting with all conditions stored is best on 27 and 33 of 60, GB-ER32 on 10 and 9, and V2 on 2 and 0.

Intervals (Table SVIII). Pooled 90% coverage is 0.89–0.91 on both datasets, but uneven by ageing type: below 0.85 on S10 calendar conditions for horizons 1–6 (0.817/0.833) and on S7 cyclic conditions (0.812–0.836), while S7 calendar samples are covered almost always (0.999). Evaluation order 0 is the table order of the conditions, in which the second half of the stream is not a random half. The calibrated ensemble of the EWC model covers 0.883/0.886 and 0.875/0.863 (S10) and 0.907/0.905 and 0.896/0.895 (S7) with intervals about 1.5 (S10) and 1.2 (S7) times as wide. No calibration cell had to borrow rows from the other origin stratum.

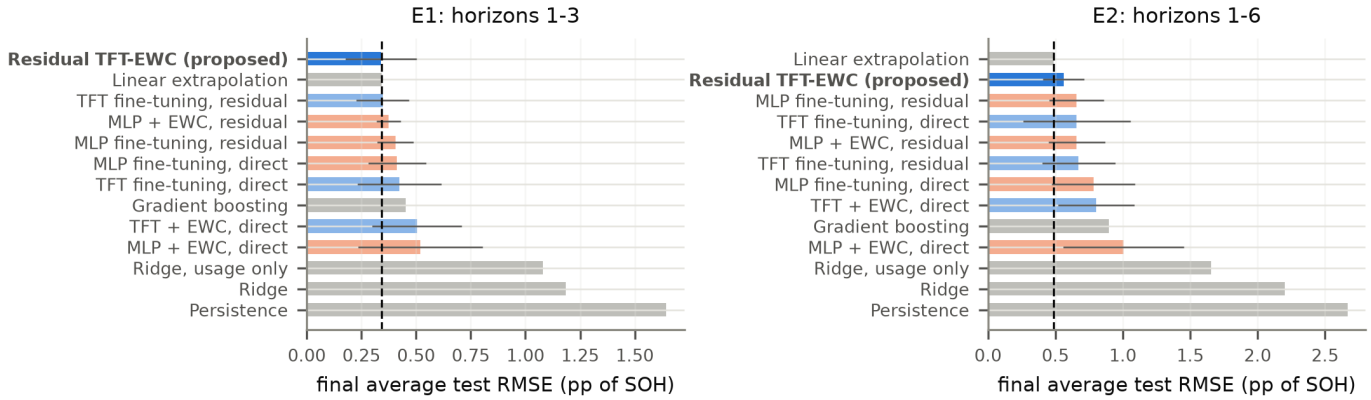


Fig. S10. S6 continual stream: final average test RMSE (pp) of every backbone, mean and s.d. over the 50 runs of the main seed set; bold, blue: the proposed residual TFT-EWC; dashed: linear extrapolation. Replicate set and forgetting: Table SIII.

TABLE SVIII

V2: INTERVAL COVERAGE, POOLED, BY AGEING TYPE AND BY EVALUATION ORDER, WITH MEAN WIDTH AND INTERVAL SCORE, MAIN/REPLICATE SEED SETS. S10 HAS NO WLTP CONDITIONS.

	S10, h=1–3	S10, h=1–6	S7, h=1–3	S7, h=1–6
90% intervals				
Pooled	0.911/0.903	0.892/0.901	0.905/0.906	0.894/0.896
Calendar	0.869/0.861	0.817/0.833	0.999/0.999	0.999/0.999
Cyclic	0.921/0.913	0.908/0.915	0.836/0.833	0.812/0.815
WLTP	–	–	0.885/0.895	0.870/0.874
Order 0	0.942/0.939	0.927/0.934	0.900/0.906	0.898/0.902
Order 1	0.907/0.897	0.898/0.893	0.900/0.902	0.895/0.898
Order 2	0.900/0.901	0.904/0.916	0.899/0.901	0.891/0.884
Order 3	0.904/0.895	0.874/0.877	0.903/0.895	0.876/0.887
Order 4	0.901/0.882	0.856/0.884	0.922/0.925	0.909/0.907
Mean width (pp)	1.28/1.22	1.91/1.91	7.12/7.26	9.61/9.80
Interval score (pp)	5.19/5.35	6.94/6.81	32.08/32.05	46.83/47.17
80% intervals				
Pooled	0.820/0.812	0.810/0.817	0.823/0.830	0.816/0.820
Calendar	0.756/0.735	0.693/0.705	0.991/0.992	0.988/0.989
Cyclic	0.836/0.831	0.835/0.842	0.720/0.726	0.707/0.714
WLTP	–	–	0.743/0.764	0.725/0.728
Order 0	0.867/0.864	0.861/0.868	0.830/0.840	0.840/0.846
Order 1	0.814/0.809	0.818/0.812	0.813/0.830	0.807/0.821
Order 2	0.807/0.807	0.826/0.843	0.797/0.810	0.810/0.793
Order 3	0.804/0.797	0.787/0.776	0.818/0.815	0.793/0.807
Order 4	0.808/0.785	0.756/0.789	0.855/0.856	0.829/0.831
Mean width (pp)	0.86/0.83	1.38/1.37	4.11/4.32	6.05/6.09
Interval score (pp)	3.13/3.22	4.33/4.25	20.49/20.41	29.75/29.97

TABLE SIX

V2 ON S10 AND S7: MEAN AND MEDIAN TEST RMSE OVER THE HELD-OUT CONDITIONS (PP), MAIN/REPLICATE SEED SETS (ONE VALUE: DETERMINISTIC METHOD). EARLIER MODEL: RESIDUAL TFT + EWC SHRUNK AT H^+ ONLY (S10) OR AT EVERY ORIGIN (S7). RESULTS/FORECASTING_D38_*.JSON.

	S10, h=1–3		S10, h=1–6		S7, h=1–3		S7, h=1–6	
Method	mean	median	mean	median	mean	median	mean	median
V2: R32 ensemble, calibrated	0.782/0.796	0.333/0.340	1.014/1.003	0.548/0.543	4.125/4.113	2.974/3.010	5.890/5.905	4.480/4.566
R32 ensemble, uncalibrated	0.782/0.794	0.333/0.333	1.013/0.998	0.543/0.535	4.124/4.086	2.954/2.981	5.868/5.871	4.547/4.560
R32, single model	0.832/0.845	0.380/0.384	1.106/1.090	0.655/0.651	4.237/4.206	3.274/3.222	6.010/6.007	4.752/4.710
R8 ensemble, calibrated	0.781/0.783	0.344/0.351	1.014/0.995	0.543/0.568	4.035/3.985	2.789/2.666	5.702/5.615	4.267/4.216
EWC model (A0), single, raw	1.110/1.103	0.634/0.650	1.592/1.598	1.206/1.215	4.787/4.707	3.421/3.382	6.824/6.711	4.908/4.996
EWC model (A0), ensemble	1.077/1.064	0.599/0.600	1.510/1.499	1.125/1.097	4.673/4.589	3.335/3.298	6.691/6.596	4.781/4.731
Earlier model (Q4 comparator)	1.040/1.038	0.552/0.568	1.542/1.552	1.134/1.115	5.201/5.184	4.106/4.103	7.489/7.481	5.831/5.803
Joint residual TFT	0.813/0.807	0.354/0.351	0.961/1.009	0.511/0.608	4.123/4.328	2.607/2.713	5.790/6.182	4.348/4.582
GB, all stored	1.388	0.353	4.773	0.832	3.845	2.706	5.117	3.721
GB-ER32	1.554	0.384	5.189	0.828	3.735	2.839	5.031	3.851
GB-ER8	1.021	0.502	1.759	1.103	4.062	3.319	5.277	4.139
GB, latest only	1.883	1.421	2.382	2.011	6.501	5.869	8.398	7.114
Linear extrapolation	1.215	0.731	1.900	1.455	5.313	4.212	7.896	6.466
Ridge	1.386	0.923	1.838	1.163	4.313	2.850	6.183	4.196
Persistence	1.392	1.121	2.181	1.779	6.605	5.593	9.108	7.950

TABLE SX

V2: REGISTERED QUESTIONS Q1–Q4. MEAN AND MEDIAN DIFFERENCE V2 MINUS COMPARATOR OVER THE HELD-OUT CONDITIONS (PP), CONDITIONS ON WHICH V2 IS BETTER, HOLM-CORRECTED WILCOXON p, MAIN/REPLICATE SEED SETS, AND THE VERDICT OF THE REPLICATION RULE. Q3 IS ON H^+ ORIGINS. A QUESTION IS ANSWERED POSITIVELY ONLY IF IT HOLDS FOR BOTH HORIZON SETS: Q1–Q4 ON S10, Q3 AND Q4 ON S7. MEAN_AND_RANK_DISAGREE IS FALSE FOR ALL SIXTEEN.

Comparison	Mean difference	Median difference	V2 better on	p_{Holm}	Verdict
S10, h=1–3 (64 conditions)					
Q1 vs. GB, all stored	−0.606/−0.592	−0.029/−0.022	42/41	0.003/0.002	better
Q2 vs. GB-ER32	−0.773/−0.758	−0.057/−0.047	50/47	$4 \cdot 10^{-5}$/$7 \cdot 10^{-5}$	better
Q3 vs. lin., H^+	−0.224/−0.204	−0.172/−0.183	56/59	$7 \cdot 10^{-8}$/$1 \cdot 10^{-7}$	better
Q4 vs. earlier model	−0.259/−0.242	−0.199/−0.190	63/62	$1 \cdot 10^{-11}$/$5 \cdot 10^{-11}$	better
S10, h=1–6 (64 conditions)					
Q1 vs. GB, all stored	−3.759/−3.770	−0.143/−0.134	47/46	$7 \cdot 10^{-6}$/$6 \cdot 10^{-6}$	better
Q2 vs. GB-ER32	−4.175/−4.185	−0.185/−0.184	57/55	$1 \cdot 10^{-8}$/$9 \cdot 10^{-9}$	better
Q3 vs. lin., H^+	−0.565/−0.569	−0.520/−0.476	56/57	$9 \cdot 10^{-9}$/$6 \cdot 10^{-9}$	better
Q4 vs. earlier model	−0.528/−0.549	−0.466/−0.491	64/64	$1 \cdot 10^{-11}$/$1 \cdot 10^{-11}$	better
S7, h=1–3 (60 conditions)					
Q1 vs. GB, all stored	+0.279/+0.267	+0.179/+0.170	17/15	0.007/0.002	worse
Q2 vs. GB-ER32	+0.390/+0.378	+0.083/+0.062	26/25	0.02/0.03	worse
Q3 vs. lin., H^+	−0.197/−0.193	−0.185/−0.163	41/43	0.02/0.009	better
Q4 vs. earlier model	−1.076/−1.071	−0.643/−0.603	55/58	$4 \cdot 10^{-10}$/$2 \cdot 10^{-10}$	better
S7, h=1–6 (60 conditions)					
Q1 vs. GB, all stored	+0.773/+0.788	+0.331/+0.421	9/9	$5 \cdot 10^{-6}$/$6 \cdot 10^{-6}$	worse
Q2 vs. GB-ER32	+0.859/+0.874	+0.418/+0.437	19/17	$1 \cdot 10^{-5}$/$2 \cdot 10^{-5}$	worse
Q3 vs. lin., H^+	−0.451/−0.380	−0.470/−0.351	43/44	0.004/0.007	better
Q4 vs. earlier model	−1.599/−1.577	−1.098/−0.921	59/59	$1 \cdot 10^{-10}$/$2 \cdot 10^{-10}$	better

TABLE SXI

V2: SECONDARY COMPARISONS AND ATTRIBUTION. MEAN DIFFERENCE OVER THE HELD-OUT CONDITIONS (PP), MAIN/REPLICATE SEED SETS, AND VERDICT (BETTER: THE FIRST-NAMED METHOD IS BETTER, REPLICATED; EQUIV.: PRACTICALLY EQUIVALENT; N.R.: NOT REPLICATED). ENSEMBLE ALONE: A0 ENSEMBLE MINUS A0 SINGLE; REPLAY ALONE: R32 SINGLE MINUS A0 SINGLE; ENSEMBLE ON R32: R32 ENSEMBLE MINUS R32 SINGLE; CALIBRATION ALONE: V2 MINUS THE UNCALIBRATED R32 ENSEMBLE; MEMORY SIZE: CALIBRATED R8 ENSEMBLE MINUS V2; R8 VS. GB-ER8: CALIBRATED R8 ENSEMBLE MINUS GB-ER8.

Comparison	S10, h=1–3	S10, h=1–6	S7, h=1–3	S7, h=1–6
V2 vs. joint residual TFT	−0.032/−0.011 equiv.	+0.054/−0.006 n.r.	+0.001/−0.215 n.r.	+0.099/−0.277 n.r.
V2 vs. GB, latest only	−1.101/−1.087 better	−1.368/−1.379 better	−2.376/−2.388 better	−2.508/−2.493 better
V2 vs. ridge	−0.604/−0.590 better	−0.824/−0.834 better	−0.189/−0.201 n.r.	−0.293/−0.278 better
V2 vs. GB, all stored, H^+	−0.715/−0.695 better	−2.640/−2.645 better	+0.024/+0.028 equiv.	+0.499/+0.570 worse
V2 vs. GB, all stored, H^-	−0.133/−0.138 n.r.	−4.714/−4.741 better	+1.051/+0.995 worse	+1.566/+1.437 worse
Ensemble alone (A0)	−0.032/−0.039 equiv.	−0.082/−0.098 better	−0.114/−0.118 better	−0.133/−0.115 better
Replay alone (R32 vs. A0, single)	−0.278/−0.258 better	−0.486/−0.508 better	−0.550/−0.501 better	−0.813/−0.704 better
Ensemble on R32	−0.050/−0.050 n.r.	−0.092/−0.093 better	−0.113/−0.120 better	−0.142/−0.135 better
Calibration alone	±0.000/+0.001 equiv.	+0.001/+0.006 equiv.	+0.001/+0.027 equiv.	+0.022/+0.033 equiv.
Memory size (R8 vs. V2)	−0.001/−0.013 equiv.	±0.000/−0.008 equiv.	−0.089/−0.128 n.r.	−0.188/−0.290 better
R8 vs. GB-ER8	−0.240/−0.239 better	−0.746/−0.764 better	−0.026/−0.076 n.r.	+0.425/+0.338 n.r.